



Grp6 Presentation

Consumer's Perception on Electric Vehicles Using Sentiment Analysis

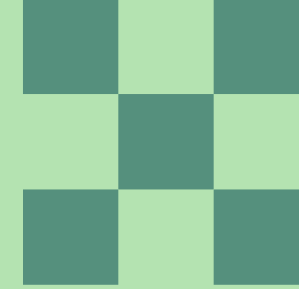


Meet the Team
SUBHAM KUMAR SAHANA





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➤ Project Purpose

Use Indian EV consumer data to run sentiment + correlation analysis, revealing drivers of choice to guide product improvements and targeted marketing

➤ Business Objective



Use Indian EV consumer data to reveal key sentiment drivers and turn them into actions for **better marketing and product fit** (top 5 drivers, campaign guidance, feature recommendations).

➤ Project objective



Data collection & integration from social, forums, reviews, and sales reports **by 15 Aug 2025**.



Sentiment pipeline implemented and validated at **≥80% accuracy by 25 Aug 2025**.



Correlation modeling linking sentiment with EV sales and technical features **by 05 Sep 2025**.

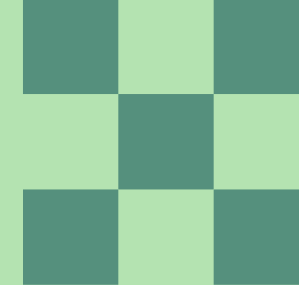


Delivery of an interactive dashboard, full documentation, and summary presentation **by 22 Sep 2025**





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➤ Deliverables

- ✓ **Interactive dashboard + summary deck** with insights, filters, and clear actions.
- ✓ **Data & code pack:** integrated 12-month dataset, ETL scripts, and reproducible workflow.
- ✓ **Model & analysis:** validated sentiment model ($\geq 80\%$ accuracy) + correlation report to sales/features.



Stakeholder Analysis



Position	Expectation	Power   	Interest	Impact   	Engagement Assessment (SEAM)	Engagement Strategy	Onion Layer
Project Sponsor	Approval & Financial support	 High	High	 High	Supportive	Regular touch-point	Layer 2 - Organization or Enterprise
Project Manager	Regular project updates, risk handling, timely completions	 High	High	 High	Leading	Daily meeting	Layer 4 - Solution Delivery
Data Engineering Lead	Execute technical tasks accurately, timely data delivery	 Moderate	High	 High	Supportive	Weekly technical meetings	Layer 4 - Solution Delivery
Sentiment Analytics Lead	Execute specific analytics tasks with accuracy	 Moderate	High	 High	Supportive	Weekly technical meetings	Layer 4 - Solution Delivery
Business Insights Analyst	Clearly defined business insights and recommendations	 Moderate	Low	 High	Neutral	Regular touch-point	Layer 3 - Affected Organizational Unit
Dashboard Developer	Complete and accurate dashboard & visual reports	 Low	Low	 Moderate	Supportive	Demo & Feedback sessions	Layer 4 - Solution Delivery
Marketing & Communication Strategist	Marketing insights, effective communication with stakeholders	 Moderate	High	 High	Leading	Weekly alignment meetings	Layer 3 - Affected Organizational Unit
Customer	Transparency, regular updates, grievance redressal, personalized experience, satisfaction, post-purchase support.	 High	High	 High	Empower/ Collaborate	Regular touchpoint	Layer 1 - Affected External stakeholders





Project Scope

✓ In Scope

- ✓ **Data acquisition:** 12-month EV sales + Twitter/Facebook/Instagram, forums, reviews
- ✓ **ETL: cleaning & preprocessing** into unified dataset
- ✓ **NLP sentiment modeling** on consumer text
- ✓ **Statistical correlation:** sentiment ↔ **sales trends/features**
- ✓ **Dashboard** for **stakeholder visualization & decisions**

✓ Out of Scope

- ✓ Primary research (surveys, focus groups)
- ✓ Real-time streaming / continuous analytics
- ✓ CRM/ERP integration
- ✓ Marketing campaign execution





Budget Summary

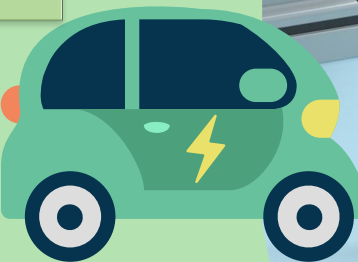
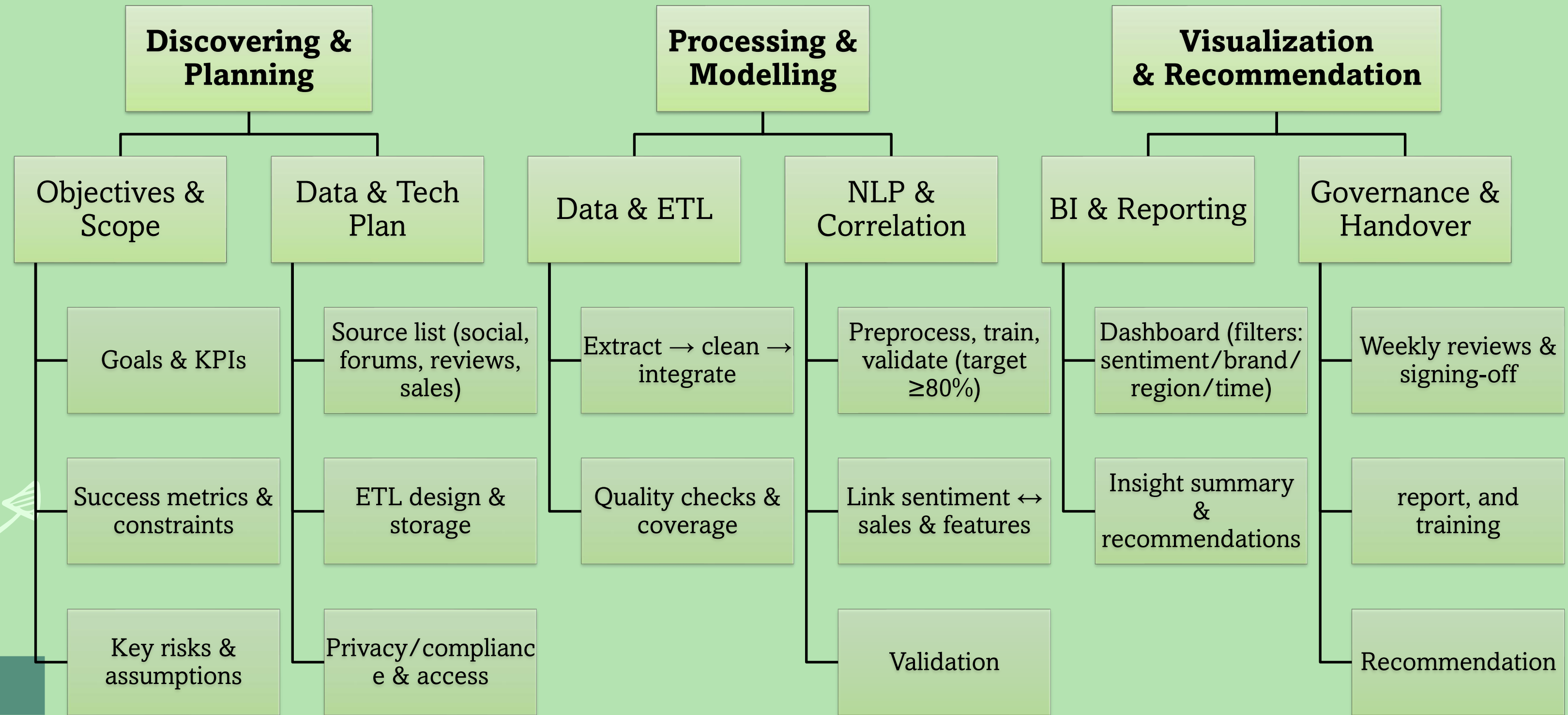
(Detailed Estimation):

Component	Estimated Cost (INR)
Personnel (Data Engineers, Data Scientists, Dashboard Developer)	3,20,000
Software Tools & Cloud Infrastructure (AWS/Azure)	90,000
Licenses (BI tools, NLP packages/API access)	30,000
Miscellaneous Expenses (travel, logistics)	20,000
Contingency Fund (10%)	86,000
Total Budget Estimate	5,46,000





Work Breakdown Structure (WBS)





PERT Analysis



Code	Activity name	Predecessor codes	Optimistic (O)	Most likely (M)	Pessimistic (P)	TE (Expected)	SD (σ)
A	Project Kick-off & Requirement Finalization	—	3	4	6	4.17	0.50
B	Data Source Access & Integration Setup	A	4	6	9	6.17	0.83
C	Data Architecture & Staging Design	A	3	4	6	4.17	0.50
D	Ingestion Pipelines (ETL/ELT) Build	B	3	5	8	5.17	0.83
E	Data Preprocessing & Quality Assurance	C, D	3	4	6	4.17	0.50
H	Sentiment Analysis Pipeline + Accuracy Validation	E	2	3	5	3.17	0.50
F	Trend Correlation & Statistical Modelling	E	3	4	7	4.33	0.67
G	Dashboard Design & Development	E	5	7	10	7.17	0.83
I	UAT, Final Presentation & Project Closure	F, G, H	1	2	3	2.00	0.33

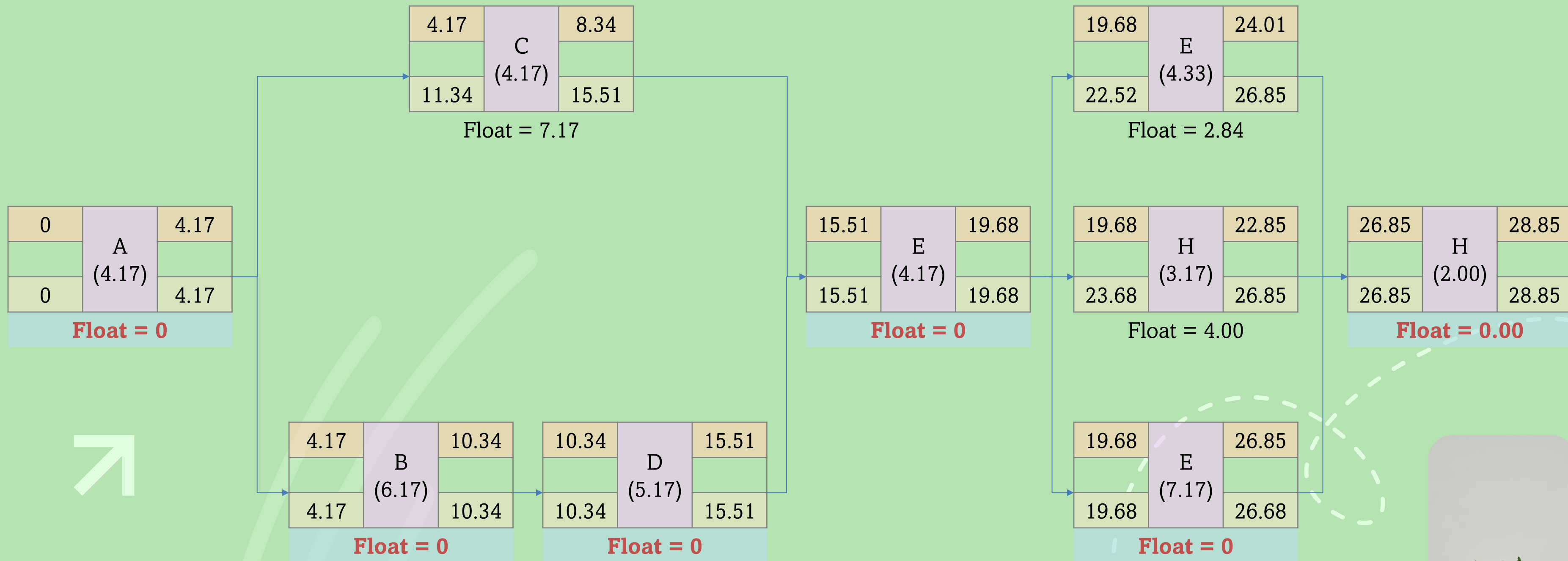
Effort = 40.52
Duration = 28.85

Standard Deviation = 1.64





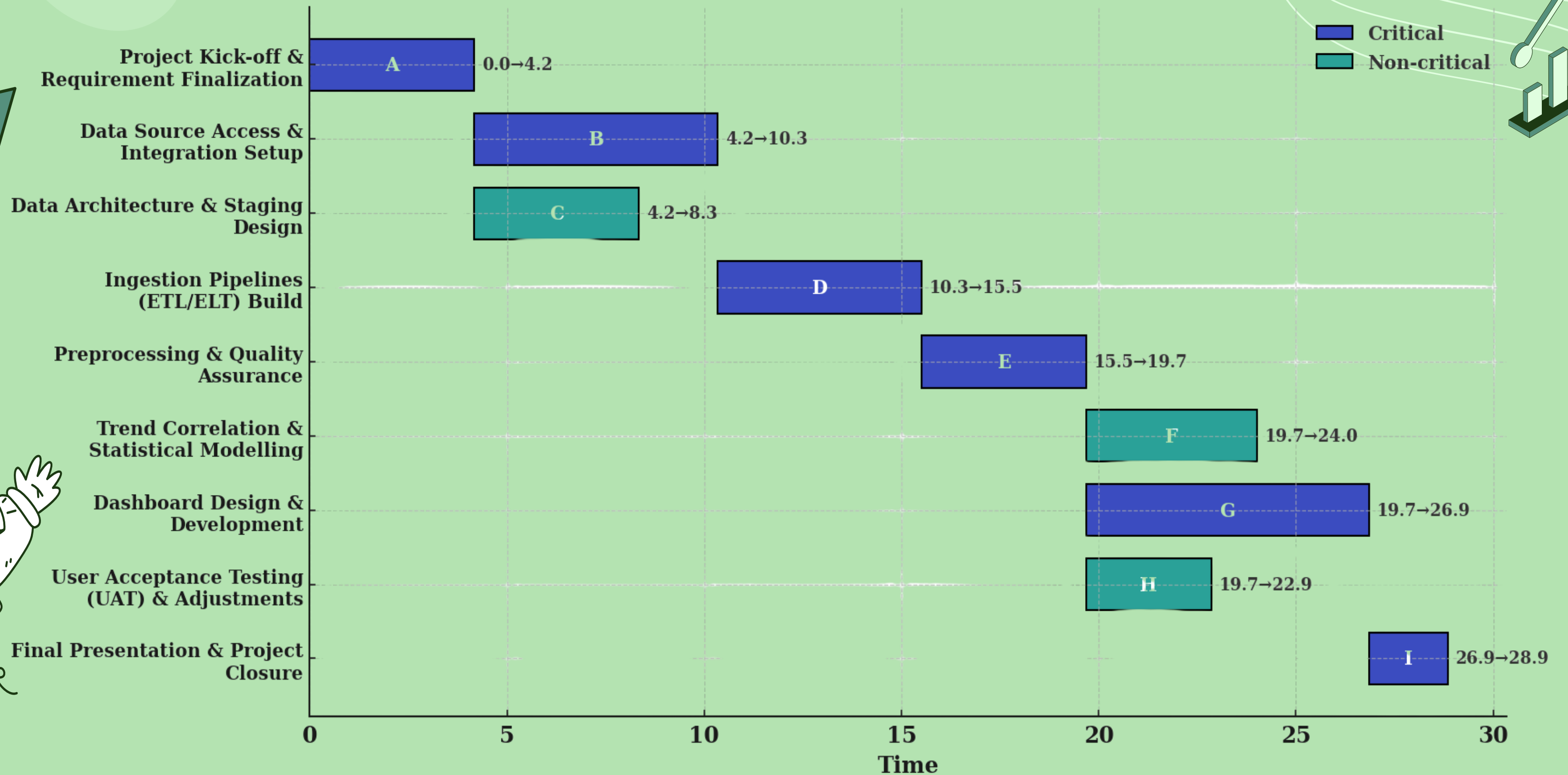
Network Diagram



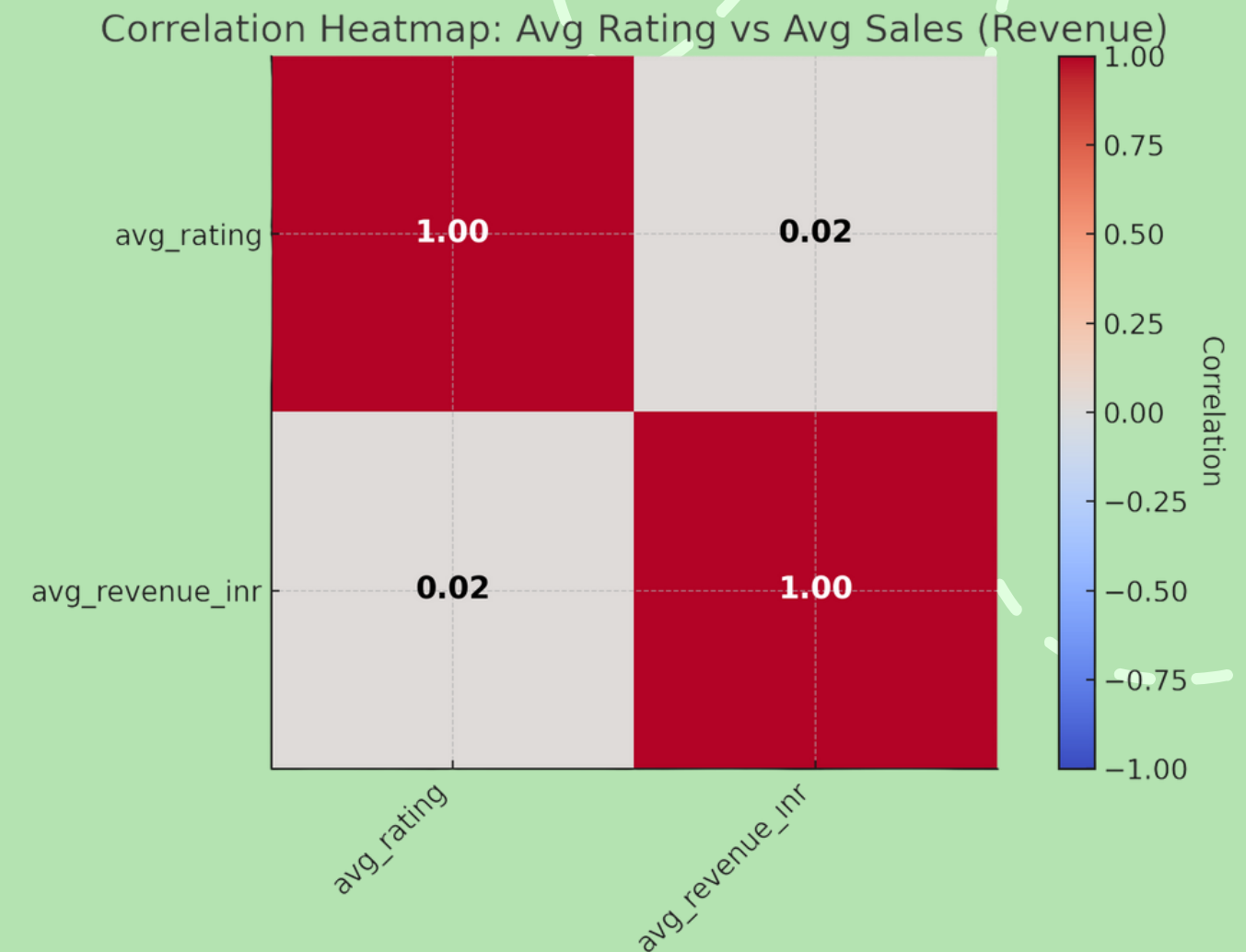
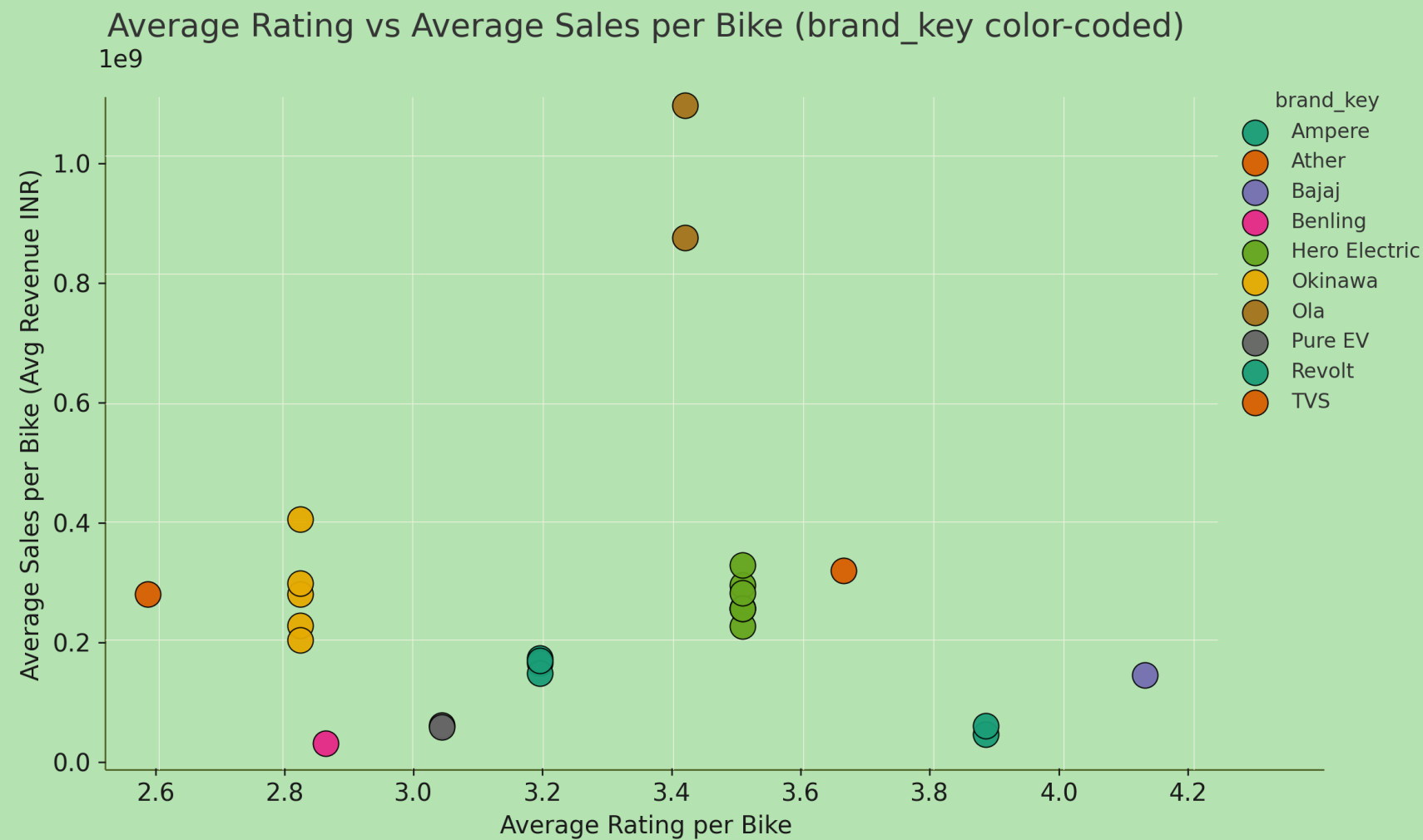
Critical Path - A → B → D → E → G → I.



Gantt Chart



correlations between consumer sentiment patterns, EV sales data



Interpretation:

- Points are widely scattered with no upward/downward trend; brand colors overlap heavily → ratings are **not a good predictor** of sales.
- Action:** Treat ratings as **supporting**, not leading, indicator for demand planning.

Interpretation:

- Sales and ratings are **almost independent** of each other.
- Higher ratings do not necessarily mean higher sales, and higher sales do not necessarily mean higher ratings.



Sentiment Analysis

Model Used: Naive Bayes classification model

=== Classification Report: Validation ===

	precision	recall	f1-score	support
negative	0.71	0.71	0.71	42
positive	0.91	0.91	0.91	127
accuracy			0.86	169
macro avg	0.81	0.81	0.81	169
weighted avg	0.86	0.86	0.86	169

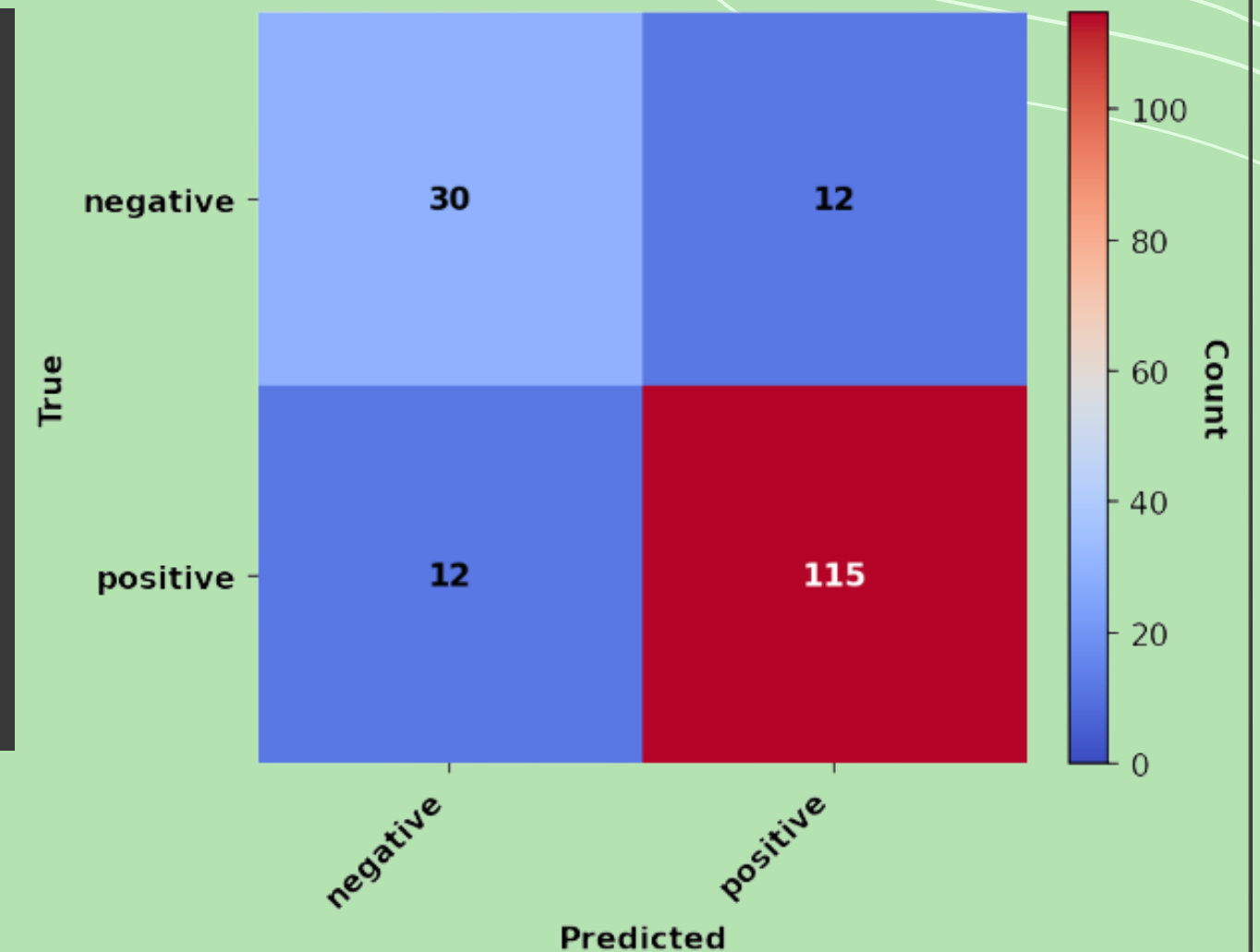
Accuracy

- test: ~90.53%
- Validation: ~ 85.80%

Interpretation (plain English)

- ✓ The model is **reliable for spotting positive sentiment** and will work well for automating “happy customer” insights, testimonials, and marketing surfaces.
- ✓ It is **less reliable at catching unhappy/critical reviews**. About **29%**(Negative miss rate (FNR for “negative”) = $12/42 \approx 28.6\%$) **of negatives are missed**, which could hide issues if you rely solely on the model to trigger support/escalation.

Validation Confusion Matrix



Dashboard

Analysis on EV-2wheeler

19

Count of Model Name

561

Count of review

3.34

Average of rating

3.55

Average of Comfort

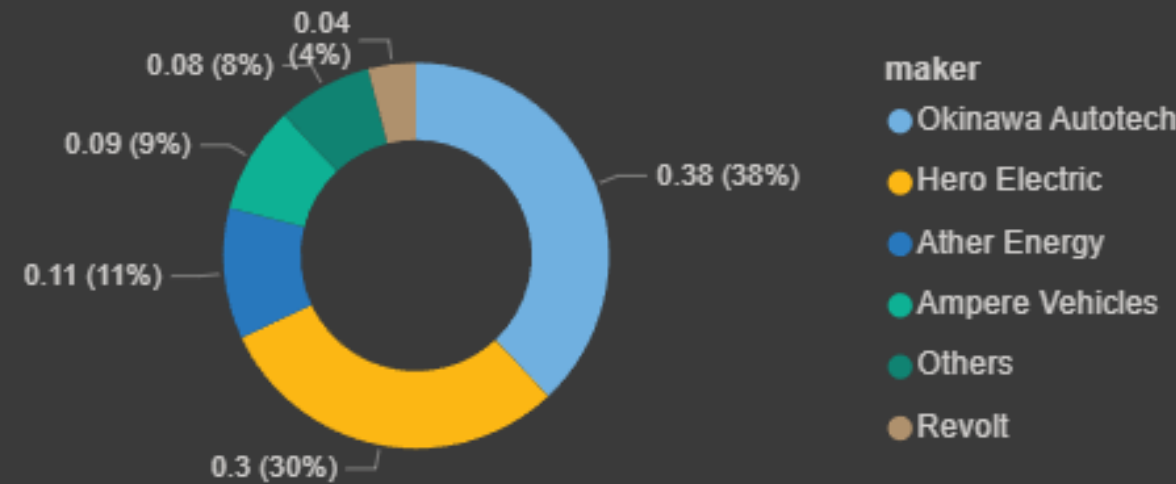
Model Name

- ☐ Ampere Magnus EX
- ☐ Ampere Magnus Pro
- ☐ Ampere REO
- ☐ Ampere Zeal
- ☐ Ather 450X
- ☐ Hero Electric Atria
- ☐ Hero Electric Flash
- ☐ Hero Electric NYX
- ☐ Hero Electric NYX ...
- ☐ Hero Electric Optima
- ☐ Hero Electric Optim...
- ☐ Hero Electric Photon
- ☐ Okinawa i-Praise
- ☐ Okinawa Lite

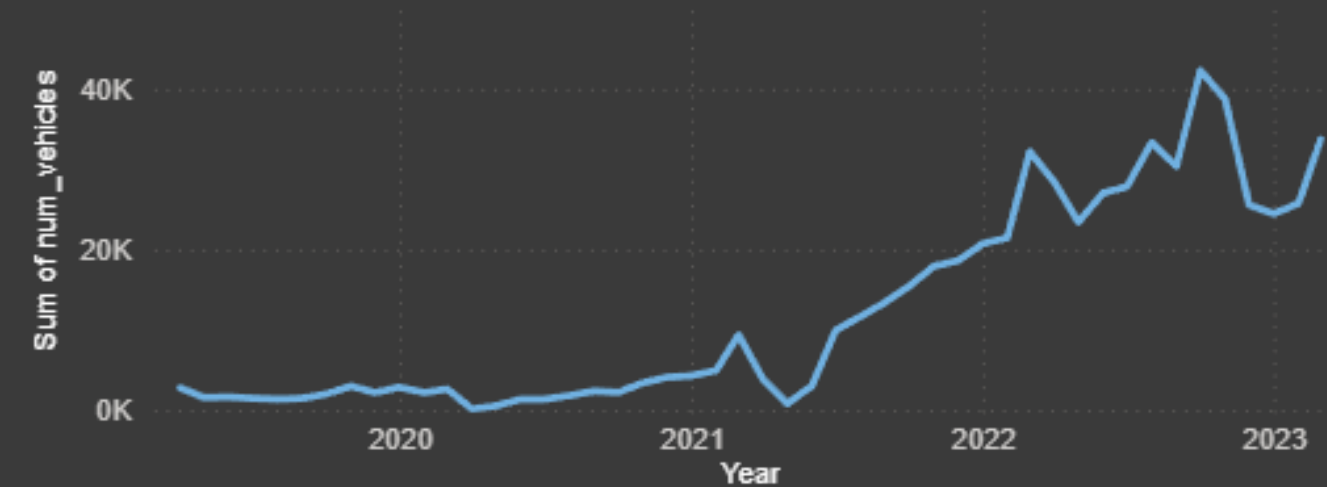
Financial_year

- ☐ 2019-20
- ☐ 2020-21
- ☐ 2021-22
- ☐ 2022-23

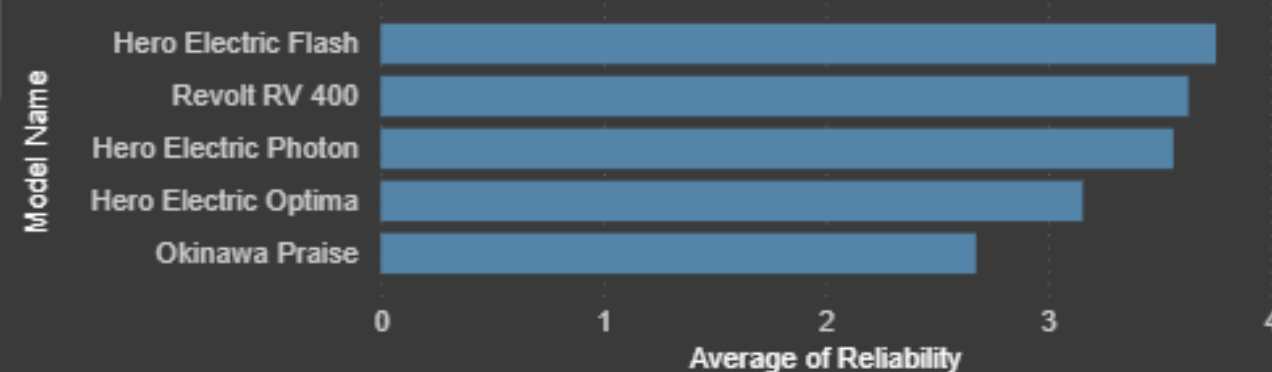
Sum of market_share by maker



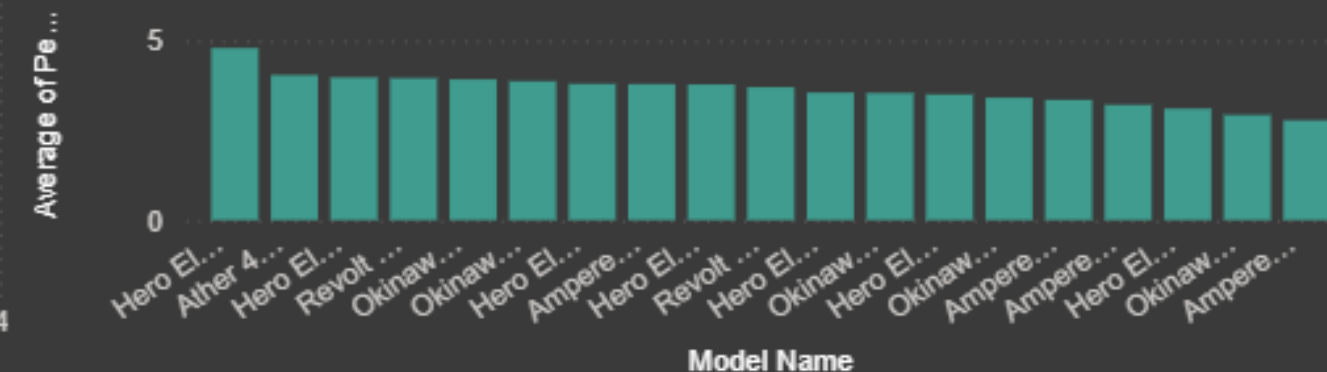
Sum of num_vehicles by Year, Quarter, Month and Day



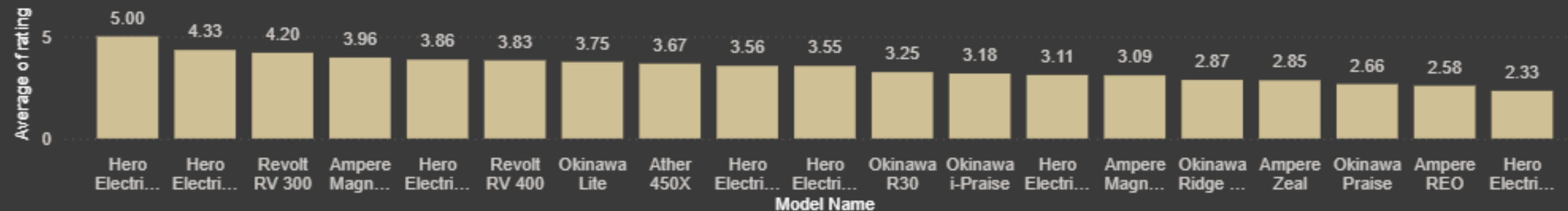
Average of Reliability by Model Name



Average of Performance by Model Name



Average of rating by Model Name





Conclusion & Suggestion

Conclusion

- ✓ We analyzed real EV consumer voice + 12-month sales and built a solid sentiment model.
- ✓ **Ratings $\neq$ sales**: helpful signal, not a predictor.
- ✓ Key drivers: **price/value, charging & range, service, build, features**.
- ✓ Dashboard gives quick, drill-down insights for action.

Suggestions

- ✓ **Marketing**: lead with *range confidence & charging ease*; show total cost benefits; use real review snippets.
- ✓ **Product**: improve charging experience, range, and service network; fix recurring build issues.
- ✓ **Ops**: alert on negative spikes; respond in 24–48 hrs; tag issues by driver.
- ✓ **Analytics**: add more sources; retrain quarterly; add aspect-based sentiment.





Thank You

Resources: [Analysis of EV 2-wheeler](#)

